

MINOR DEGREE IN ARTIFICIAL INTELLIGENCE

24AI007	DATA SCIENCE FOR ENGINEERS	L	T	P	C
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OBJECTIVES: The Course will enable learners to: • To learn the fundamentals of Data Science. • To describe and understand data for analysis. • To describe and analyze relationships between variables. • To apply various data pre-processing strategies. • To apply data wrangling on data for further processing.					
UNIT I	INTRODUCTION				9
Data Science: Benefits and uses – facets of data - Data Science Process: Overview – Defining research goals – Retrieving data – data preparation - Exploratory Data analysis – build the model – presenting findings and building applications - Data Mining - Data Warehousing – Basic statistical descriptions of Data.					
UNIT II	DESCRIBING DATA				9
Types of Data - Types of Variables -Describing Data with Tables and Graphs –Describing Data with Averages - Describing Variability - Normal Distributions and Standard (z) Scores.					
UNIT III	DESCRIBING RELATIONSHIPS				9
Correlation –Scatter plots –correlation coefficient for quantitative data –computational formula for correlation coefficient – Regression –regression line –least squares regression line – Standard error of estimate – interpretation of r2 –multiple regression equations –regression towards the mean.					
UNIT IV	DATA PREPROCESSING				9
Data Preprocessing – Purpose – Tools – Levels – Cleaning up the Table – Unpacking – Restructuring – Reformulating the Table - Data Cleaning: – Missing Values – Outliers – Errors - Data Fusion vs Data Integration – Directions – Adding attributes – data objects - Entity identification – Data Reduction vs data redundancy – Types – Performing Numerosity Data reduction – Sampling – Principal Component Analysis – Data Transformation – Normalization and Standardization – Discretization – Smoothing – Aggregation - Binning.					
UNIT V	DATA WRANGLING				9
Processing Uni-dimensional Data: Creating Vectors - Inspecting the Data Distribution with Histograms- Aggregating Numerical Data - Arithmetic Operators - Indexing vectors- Multidimensional data: Creating matrices - Reshaping - visualization (2d,3d) - Heterogeneous Data: Data frames - Aggregation - Transformation - Indexing - Accessing Database: Filtering - Ordering - Removing Duplicates - Grouping and Aggregating - Joining - Handling with Many Files.					
TOTAL:45 PERIODS					
OUTCOMES: At the end of this course, the students will be able to: CO1: Explain the fundamentals of data science. CO2: Illustrate the basics of data for analysis. CO3: Identify relationships between variables. CO4: Implement various data pre-processing strategies. CO5: Implement data wrangling for further processing of data. CO6: Apply Data preprocessing on real-time data sets.					
TEXT BOOKS:					

1. Robert S. Witte and John S. Witte, “Statistics”, Eleventh Edition, Wiley Publications, 2017. (Units 2, 3)
2. Roy Jafari, Hands-On Data Preprocessing in Python: Learn how to effectively prepare data for successful data analytics, Packt Publications, First Edition, 2022. (Unit 4)
3. Marek Gagolewski, Minimalist Data Wrangling with Python, Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND 4.0), 2022. (Unit 5)
REFERENCES:
1. David Cielien, Arno D. B. Meysman, and Mohamed Ali, “Introducing Data Science”, Manning Publications, 2016. (Unit 1)
2. Hui Lin, Ming Li, Practitioner’s Guide to Data Science, 1st Edition, Chapman and Hall/CRC, 2023.
3. Jiawei Han, Micheline Kamber, Jian Pei, "Data Mining: Concepts and Techniques", 3rd Edition, Morgan Kaufmann, 2012.
4. Suresh Kumar Mukhiya, Usman Ahmed, Hands-On Exploratory Data Analysis with Python: Perform EDA techniques to understand, summarize, and investigate your data, Packt Publications, 2020.
5. Dr. Tirthajyoti Sarkar, Shubhadeep Roychowdhury, Data Wrangling with Python: Creating actionable data from raw sources, Packt Publications, First Edition, 2019.
6. Robert H. Shumway, David S. Stoffer, Time Series Analysis and Its Applications With R Examples, Fourth Edition, 2016, Springer.
7. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements Of Statistical Learning: Data Mining, Inference, and Prediction, Springer, Second Edition, 2017.
8. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, O'Reilly, Third Edition, 2022.
9. Jake VanderPlas, “Python Data Science Handbook – Essential tools for working with data”, O’Reilly, 2017.
10. Ashwin Pajankar, Aditya Joshi, Hands-on Machine Learning with Python: Implement Neural Network Solutions with Scikit-learn and PyTorch, Apress, 2022.
11. NPTEL Courses: a. Data Science for Engineers - https://onlinecourses.nptel.ac.in/noc23_cs17/preview b. Python for Data Science - https://onlinecourses.nptel.ac.in/noc23_cs21/preview